

Functional Design Document (FDD)

Project: SearchFusion – Enterprise Saved Search Merge & Reporting Platform (NetSuite)

Version: 1.0 **Date:** 09 Mar 2026 **Owner:** Kishor Patil

1. Purpose

SearchFusion provides a generic enterprise framework to merge results from two or more NetSuite Saved Searches into a unified dataset, configure it through a UI, export it (CSV/Excel/PDF), schedule distribution, and optionally use AI (third-party LLM) for mapping/summaries. It can also support SuiteAnalytics Workbook via optional data materialization.

2. Background / Business Problem

Teams maintain multiple saved searches across different transaction types and operational areas. Stakeholders need consolidated, standardized reporting with one-click export and automated recurring distribution. Manual consolidation is time-consuming, inconsistent, and increases audit risk.

Current limitations: Saved searches are not naturally mergeable across different sources/types; standard exports are limited per search; manual consolidation leads to inconsistency and dependency on technical resources.

3. Scope

3.1 In Scope

- Create configurable Report Definitions.
- Add N saved searches as Sources for a report.
- Define standardized Output Columns (schema).
- Map each Source's columns into Output Columns (label/internal ID mapping modes).
- Union merge (append) across sources; optional dedupe by key(s); optional sort by key(s).
- Runtime filtering at execution time (NetSuite filters JSON).
- Preview merged results with pagination.
- Exports: CSV (full, multi-part), Excel (SpreadsheetML; may be multi-part), PDF summary (totals/anomalies/sample).
- Scheduling (daily/weekly/monthly) with idempotency + locking to avoid duplicate runs.
- Run History with audit trail and downloadable file retention.
- AI integration (optional): auto-mapping suggestions, run summaries, anomaly explanation text.
- Optional SuiteAnalytics Workbook enablement via materialized dataset.

3.2 Out of Scope (Phase 2 / Optional)

- Join-based merging (inner/left joins across searches by key).
- True streaming into a single huge Excel workbook.

- Full row-level PDF export for large datasets.
- Full lineage visualization and versioned mapping history UI (beyond run history).
- Full materialization pipeline at massive scale (if enabled later).

4. Stakeholders & User Roles

4.1 Stakeholders

- Finance leadership (report consumers).
- Operations leadership (scheduled reporting).
- NetSuite Admin (deployment + governance).
- Technical team (SuiteScript / integrations / support).

4.2 User Roles (RBAC)

- **Admin:** Create/edit all reports, sources, mappings, schedules; manage AI settings and access.
- **Editor:** Maintain report definitions within allowed scope.
- **Viewer:** View/run reports, preview results, export.
- **Scheduler (optional):** Manage schedules and recipients only.

RBAC is controlled per report using “Allowed Roles (View/Edit)” settings and enforced server-side.

5. Key Definitions

- **Report Definition:** Configuration container for a consolidated report.
- **Source:** A saved search included in a report.
- **Output Column (Schema):** Standardized field definition (key, label, type).
- **Mapping:** How a source column populates an output column.
- **Run:** Execution instance that produces preview/export outputs.
- **Materialization:** Persisting merged rows in a custom record for analytics/workbooks.

6. User Journeys

- **Create Report:** User creates a report, configures preview/sort/dedupe/default output/RBAC, and saves.
- **Add Sources:** User adds N saved searches as sources; enables/disables; selects mapping mode (LABEL or INTERNAL_ID).
- **Define Output Schema:** User defines output columns (key/label/type/transform/required/default).
- **Map Schema:** User maps output columns to source columns; optionally uses AI auto-map; saves mappings.
- **Preview & Validate:** User runs preview with optional runtime filters; reviews totals and anomalies.
- **Export Output:** User exports CSV/Excel/PDF; system runs Map/Reduce and stores files in run history; optional email distribution.

- **Schedule Automatic Runs:** User defines schedules; scheduler triggers exports and logs runs.

7. Functional Requirements

7.1 Report Definition

- Create, edit, delete a report.
- Properties: name, default output, preview cap, sort keys, dedupe key, AI toggle, allowed roles view/edit.

7.2 Source Management

- Add/edit/delete sources.
- Fields: saved search id, name, enabled flag, category/tag, mapping mode (LABEL/INTERNAL_ID).

7.3 Output Schema

- Add/edit/delete output columns.
- Fields: key (unique), label, type (text/number/currency/date), required, default, transform (trim/toNumber).

7.4 Mapping

- Save mapping grid per source per output key.
- Fallback matching by label when explicit mapping is missing.
- Use defaults where configured; required validation produces anomalies.

7.5 Merge Execution Logic

- Load each enabled saved search, apply runtime filters (optional), normalize rows into output schema.
- Union merge results, optional dedupe and sort.
- Preview uses a configurable cap; exports handle full volume via Map/Reduce.

7.6 Validation & Anomalies

- Detect missing required fields, negative numeric values, invalid totals, duplicate keys (if configured).
- Store anomalies JSON in run history.

7.7 Exports

- CSV: full export via multi-part output files.
- Excel: SpreadsheetML export (may be multi-part).
- PDF: summary PDF including totals/anomalies/sample rows.
- Store produced file IDs in run history.

7.8 Run History & Audit

- Run record captures status, trigger (UI/SCHEDULE), row counts, file list, totals, anomalies, runtime filters, errors.

7.9 Scheduling

- Daily/weekly/monthly schedules with due time window matching.
- Idempotency key prevents duplicate runs; lock prevents parallel triggers.
- Scheduled runs submit Map/Reduce exports and create run history.

7.10 AI Integration (Optional)

- Auto map suggests outputKey → source column labels.
- AI summary creates short executive narrative from totals/anomalies/deltas.
- All AI calls routed through a secure gateway; keys never exposed to UI.

8. Non-Functional Requirements

- **Security:** RBAC enforced server side; Suitelet access controlled by deployment audience; AI keys secured via gateway.
- **Performance:** Preview cap; exports via Map/Reduce; multi-part outputs; optional materialization for very large datasets.
- **Reliability:** Idempotency + lock on schedules; failures captured in run history with error details.
- **Auditability:** Run history retains inputs (filters/config) and outputs (file IDs).

9. Data Model (Custom Records)

- **Report:** name, default output, preview cap, sort keys, dedupe key, AI enabled, allowed roles view/edit.
- **Source:** report ref, saved search id, name, enabled, mapping mode.
- **Output Column:** report ref, key/label/type/required/default/transform.
- **Mapping:** report ref, source ref, outputKey, label/internal id.
- **Schedule:** report ref, frequency/time/byday/dom, output format, recipients, enabled, last_run_key, lock.
- **Run:** report ref, status/trigger, row count, files JSON, totals JSON, anomalies JSON, filters JSON, error text.

10. UI Requirements

- **Screens:** Reports List; Report Builder (Settings/Sources/Columns/Mappings/Schedules); Run (Preview/Export); Run History (download files).
- **Usability:** Non-technical friendly mapping; validations inline; export status visible via run history.

11. Acceptance Criteria

- User can merge 3+ saved searches into one output.
- User can define unified schema and map source columns.

- Preview shows merged rows and anomalies.
- Export generates downloadable files and records run history.
- Scheduler triggers runs correctly without duplicates.
- Failures are captured and visible in run history.

12. Risks & Mitigations

- **Saved search labels change** → use INTERNAL_ID mapping where possible; add mapping validation checks.
- **Excel scaling limits** → prefer CSV for large exports; multi-part Excel; consider materialization + workbook dashboards.
- **Scheduling timezones** → schedule in account time with buffer windows; document constraints.
- **AI hallucination** → AI used for suggestions/summaries only; never a source of truth.

13. Future Enhancements (Roadmap)

- Join-based merging (key-based inner/left joins).
- Full materialize pipeline and workbook templates.
- Field-level lineage and mapping validation UI.
- Threshold-based alerts (email/slack) when anomalies exceed limits.
- Governance features for large multi-team deployments.